CHAPTER 4

Vector Spaces

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Chapter 4 Vector Spaces（向量空间）

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§ 4.1 Vector Spaces and Subspaces

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§ 4.1 Vector Spaces and Subspaces

Vector Spaces

Definition： A vector space is a **nonempty** set V of objects, called *vectors*, on which are defined two operations, called ***addition*** and ***multiplication by scalars*** (real numbers), subject to the ten axioms below. The axioms must hold for all **u**, **v** and **w** in V and for all scalars c and d：

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(1) **u + v** is in V.

(2) c**u** is in V.

有时这两条不显示地写出来

(3) **u + v =v + u**.

(4) **(u + v) + w = u + v + w).**

(5) There is a zero vector **0** in V such that **u + 0 = u**（存在零元）.

(6) For each **u** in V, there is vector w in V such that

u+ 0.(存在负元或逆元）

(7) c(**u** + **v**) c**u** + c**v**. (8) $(c + d)$**u** = c**u** + d**u**.

(9) $c(d$**u**) = （ cd)**u**. (10) 1**u** = **u**.

- The following can be proved(not difficult):

(1) The zero vector **0** of a vector space V is unique.

(2) For each v in V, its negative is unique, dented by –u, and we have –u=(-1)u.

(3) For each u in V and scalar 0, we have 0u= **0.**

(4) For **0** in *V and any scalar k, we have k* **0= 0**

- Example：R^n，where n ≥ 1is a vector space.

EXAMPLE: Let $M_{2\times 2} = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \text{ are real} \right\}$

Then $M_{2\times 2}$ is a vector space.

In this context, note that the **0** vector is $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

EXAMPLE: Let $n \geq 0$ be an integer and let

$$\mathbf{P}_n = \text{the set of all polynomials of degree at most } n \geq 0.$$

Members of $\mathbf{P}_n$ have the form

$$\mathbf{p}(t) = a_0 + a_1 t + a_2 t^2 + \cdots + a_n t^n$$

where $a_0, a_1, \ldots, a_n$ are real numbers and t is a real variable. The set $\mathbf{P}_n$ is a vector space.

Let $\mathbf{p}(t) = a_0 + a_1 t + \cdots + a_n t^n$ and $\mathbf{q}(t) = b_0 + b_1 t + \cdots + b_n t^n$. Let c be a scalar.

***Axiom* 1:**

The polynomial $\mathbf{p} + \mathbf{q}$ is defined as follows:

$(\mathbf{p} + \mathbf{q})(t) = \mathbf{p}(t) + \mathbf{q}(t)$. Therefore,

$$(\mathbf{p} + \mathbf{q})(t) = \mathbf{p}(t) + \mathbf{q}(t)$$

$$= (\underline{\quad a_0 + b_0 \quad}) + (\underline{\quad a_1 + b_1 \quad})t + \cdots + (\underline{\quad a_n + b_n \quad})t^n$$

which is also a ____polynomial____ of degree at most ____n____. So $\mathbf{p} + \mathbf{q}$ is in $\mathbf{P}_n$.

***Axiom* 4:**

$$\mathbf{0} = 0 + 0t + \cdots + 0t^n$$

(zero vector in $\mathbf{P}_n$)

$$(\mathbf{p} + \mathbf{0})(t) = \mathbf{p}(t) + \mathbf{0} = (a_0 + 0) + (a_1 + 0)t + \cdots + (a_n + 0)t^n$$

$$= a_0 + a_1 t + \cdots + a_n t^n = \mathbf{p}(t)$$

and so $\mathbf{p} + \mathbf{0} = \mathbf{p}$

***Axiom* 6:**

$$(c\mathbf{p})(t) = c\mathbf{p}(t) = (___c\,a_0___) + (___c\,a_1___)t + \cdots + (___c\,a_n___)t^n$$

which is in $\mathbf{P}_n$.

The other 7 axioms also hold, so $\mathbf{P}_n$ is a vector space.

§ 4.1 Vector Spaces and Subspaces

Subspaces

Definition： A **subspace** of a vector space V is a subset H of V that has three properties:

a. The zero vector of V is in H.

b. H is closed under vector addition. That is, for each **u** and **v** are in H, **u + v** is in H.

c. H is closed under scalar multiplication. That is, for each **u** in H and each scalar c, c**u** is in H.

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- Example ： The set consisting of only the zero vector in a vector space of V, called the zero subspace and written as {**0**}.

- Example ： The vector space R^2 is not a subspace of R^3 because R^2 is not even a subset of R^3. The set

$$H = \left\{ \begin{bmatrix} s \\ t \\ 0 \end{bmatrix} : s,t\text{是实数} \right\}$$

is a subset of R^3 that “looks” and “acts” like R^2, although it is logically distinct from R^2. *H* is a subspace of R^3

A Subspace Spanned by a Set

Example : Given $\mathbf{v}_1$ and $\mathbf{v}_2$ in a vector space V, let H = span $\{\mathbf{v}_1, \mathbf{v}_2\}$. Show that H is a subspace of V.

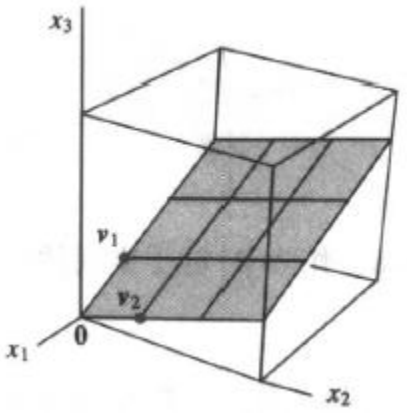
Solution : The zero vector is in H, since $\mathbf{0} = 0\mathbf{v}_1 + 0\mathbf{v}_2$.

To show that H is closed under vector addition, take two arbitrary vectors in H, $u = s_1\mathbf{v}_1 + s_2\mathbf{v}_2$, $w = t_1\mathbf{v}_1 + t_2\mathbf{v}_2$

and

$$u + w = (s_1\mathbf{v}_1 + s_2\mathbf{v}_2) + (t_1\mathbf{v}_1 + t_2\mathbf{v}_2)$$
$$= (s_1 + t_1)\mathbf{v}_1 + (s_2 + t_2)\mathbf{v}_2$$

$$cu = c(s_1\mathbf{v}_1 + s_2\mathbf{v}_2) = (cs_1)\mathbf{v}_1 + (cs_2)\mathbf{v}_2$$



So u + w is in H, and cu is in H. Thus H is a subspace of V.

§ 4.1 Vector Spaces and Subspaces

Theorem 1

If $v_1,\ldots,v_p$ are in a vector space V, then $\text{span}\{v_1,\ldots,v_p\}$ is a subspace of V.

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Proof: In order to verify this, check properties a, b and c of definition of a subspace.

a. $\mathbf{0}$ is in $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ since

$$\mathbf{0} = \underline{0}\,\mathbf{v}_1 + \underline{0}\,\mathbf{v}_2 + \cdots + \underline{0}\,\mathbf{v}_p$$

b. To show that $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ closed under vector addition, we choose two arbitrary vectors in $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$:

$$\mathbf{u} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_p\mathbf{v}_p$$

and

$$\mathbf{v} = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \cdots + b_p\mathbf{v}_p.$$

Then

$$\mathbf{u} + \mathbf{v} = (a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_p\mathbf{v}_p) + (b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \cdots + b_p\mathbf{v}_p)$$

$$= (\underline{a_1}\,\mathbf{v}_1 + \underline{b_1}\,\mathbf{v}_1) + (\underline{a_2}\,\mathbf{v}_2 + \underline{b_2}\,\mathbf{v}_2) + \cdots + (\underline{a_p}\,\mathbf{v}_p + \underline{b_p}\,\mathbf{v}_p)$$

$$= (a_1 + b_1)\mathbf{v}_1 + (a_2 + b_2)\mathbf{v}_2 + \cdots + (a_p + b_p)\mathbf{v}_p.$$

So $\mathbf{u} + \mathbf{v}$ is in $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$.

c. To show that Span$\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ closed under scalar multiplication, choose an arbitrary number c and an arbitrary vector in Span$\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$:

$$\mathbf{v} = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \cdots + b_p\mathbf{v}_p.$$

Then

$$c\mathbf{v} = c(b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \cdots + b_p\mathbf{v}_p)$$

$$= \underline{c\,b_1}\,\mathbf{v}_1 + \underline{c\,b_2}\,\mathbf{v}_2 + \cdots + \underline{c\,b_p}\,\mathbf{v}_p$$

So $c\mathbf{v}$ is in Span$\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$.

Since properties a, b and c hold, Span$\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is a subspace of V.

- Example ： Let *H* be the set of all vectors of the form (a-3b, b-a, a, b), where a and b are arbitrary scalars. That is,

$$H = \{(a-3b, b-a, a, b) : a, b \in \mathbb{R}\}$$

Show that *H* is a subspace of R^4.

- Solution ： Write the vectors in H as column vectors. Then an arbitrary vector in H has the form:

$$\begin{bmatrix} a-3b \\ b-a \\ a \\ b \end{bmatrix} = a\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

v_1 v_2

This calculation shows that H = Span{v1,v2}.

Thus *H* is a subspace of R^4

EXAMPLE: Is $H = \left\{ \begin{bmatrix} a+2b \\ a+1 \\ a \end{bmatrix} : a \text{ and } b \text{ are real} \right\}$ a subspace of $\mathbf{R}^3$? Why or why not?

EXAMPLE: Is the set H of all matrices of the form $\begin{bmatrix} 2a & b \\ 3a+b & 3b \end{bmatrix}$ a subspace of $M_{2\times 2}$? Explain.

Solution: Since

$$\begin{bmatrix} 2a & b \\ 3a+b & 3b \end{bmatrix} = \begin{bmatrix} 2a & 0 \\ 3a & 0 \end{bmatrix} + \begin{bmatrix} 0 & b \\ b & 3b \end{bmatrix}$$

$$= a\begin{bmatrix} 2 & 0 \\ 3 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}.$$

Therefore H =Span $\left\{\begin{bmatrix} 2 & 0 \\ 3 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}\right\}$ and so H is a subspace of $M_{2\times 2}$.

Assignment

- Ex 4.1: 2, 8,11,16,18

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§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

- 1. The Null Space of a Matrix
- 2. The Column Space of a Matrix
- 3. The Contrast Between Nul A and Col A
- 4. Kernel and Range of a Linear Transformation

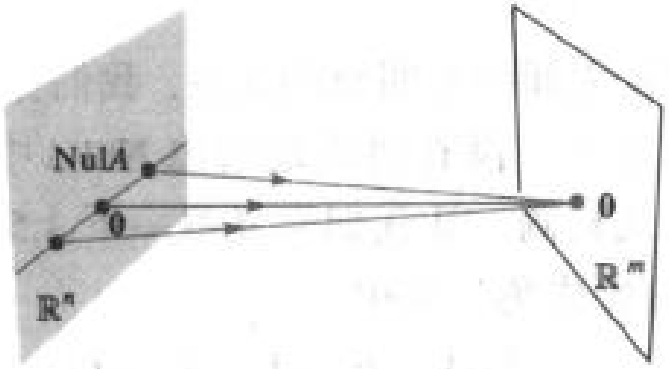
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§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

The Null Space of a Matrix (零空间)

The **null space** of an $m \times n$ matrix A, written as Nul A, is the set of all solutions to the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

$$\text{Nul } A = \{\mathbf{x} : \mathbf{x} \text{ is in } \mathbf{R}^n \text{ and } A\mathbf{x} = \mathbf{0}\} \qquad \text{(set notation)}$$



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- Example： Let $A=\begin{bmatrix}1 & -3 & -2\\ -5 & 9 & 1\end{bmatrix}$ $u=\begin{bmatrix}5\\ 3\\ -2\end{bmatrix}$

  Determine if u belongs to the null space of A.

- Solution： To test if u satisfies Au = 0, simply compute

$$Au=\begin{bmatrix}1 & -3 & -2\\ -5 & 9 & 1\end{bmatrix}\begin{bmatrix}5\\ 3\\ -2\end{bmatrix}=\begin{bmatrix}5-9+4\\ -25+27-2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

  Thus u is in Nul A.

§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

THEOREM 2

The null space of an $m \times n$ matrix A is a subspace of $\mathbf{R}^n$. Equivalently, the set of all solutions to a system $A\mathbf{x} = \mathbf{0}$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbf{R}^n$.

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- Proof: Nul A is a subset of R^n since A has n columns. Must verify properties a, b and c of the definition of a subspace.

- Property (a) Show that 0 is in Nul A. Since A0 = 0, 0 is in R^n.
- Property (b) If u and v are in Nul A, show that u + v is in Nul A. Since u and v are in Nul A, Au = 0 and Av = 0. Therefore:

$$A(\mathbf{u}+\mathbf{v}) = \underline{\text{Au}} + \underline{\text{Av}} = \underline{0} + \underline{0} = \underline{0}.$$

- Property (c) If u is in Nul A and c is a scalar, show that cu in Nul A:

$$A(c\mathbf{u}) = \underline{\subset} A(\mathbf{u}) = c\mathbf{0} = \mathbf{0}.$$

- Since properties a, b and c hold, A is a subspace of R^n.

- Example ： Let *H* be the set of all vectors in R^4 whose coordinates a, b, c, d satisfy the equation:

$$a-2b+5c=d \qquad c-a=b$$

Show that *H* is a subspace of R^4.

- Solution： H is the set of all solutions of the following system of homogeneous linear equation:

$$a-2b+5c-d=0$$
$$-a-b+c=0$$

- By Theorem 2, H is a subspace of R^4.

EXAMPLE: Find an explicit description of Nul A where

$$A = \begin{bmatrix} 3 & 6 & 6 & 3 & 9 \\ 6 & 12 & 13 & 0 & 3 \end{bmatrix}$$

Solution: Row reduce augmented matrix corresponding to $A\mathbf{x} = \mathbf{0}$:

$$\begin{bmatrix} 3 & 6 & 6 & 3 & 9 & 0 \\ 6 & 12 & 13 & 0 & 3 & 0 \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 & 2 & 0 & 13 & 33 & 0 \\ 0 & 0 & 1 & -6 & -15 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2x_2 - 13x_4 - 33x_5 \\ x_2 \\ 6x_4 + 15x_5 \\ x_4 \\ x_5 \end{bmatrix}$$

$$= x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -13 \\ 0 \\ 6 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -33 \\ 0 \\ 15 \\ 0 \\ 1 \end{bmatrix}$$

Then

$$\text{Nul } A = \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$$

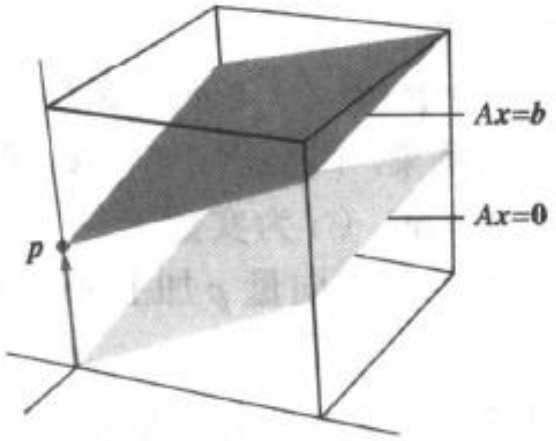
补充一些线性方程组解的结构

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§ 1.5 Solution Set of Linear Systems

- **Theorem 6（ 解的结构 ）(p.63)**

Suppose the equation Ax = b is consistent for some given b, and let p be a solution. Then the solution set of Ax = b is the set of all vectors of the form w= p + v_h, where v_h is any solution of the homogeneous equation Ax = 0.



非齐通解=非齐特解+齐通解

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§4.2 Null Spaces
齐次线性方程组解的结构

THEOREM 2

The null space of an $m \times n$ matrix A is a subspace of $\mathbf{R}^n$. Equivalently, the set of all solutions to a system $A\mathbf{x} = \mathbf{0}$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbf{R}^n$.

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线性方程组解的结构

- $AX = b$, 有解的情况下:
- $AX = b$ *的一个特解*
- 对应齐次线性方程组 $AX = 0$ 的通解（ 一般解 ）

齐次线性方程组的解集的一个极大无关组就称为该齐次线性方程组的一个 ***基础解系***

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§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

The Column Space of a Matrix

Definition：

The **column space** of an $m \times n$ matrix A (Col A) is the set of all linear combinations of the columns of A.

If $A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}$, then

$$\text{Col } A = \text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$$

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§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

Theorem 3

The column space of an $m \times n$ matrix A is a subspace of $\mathbf{R}^m$.

Recall that if $A\mathbf{x} = \mathbf{b}$, then $\mathbf{b}$ is a linear combination of the columns of A. Therefore

$$\text{Col } A = \{\mathbf{b} : \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \text{ in } \mathbf{R}^n\}$$

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- Example : Find a matrix A such that W = Col A.

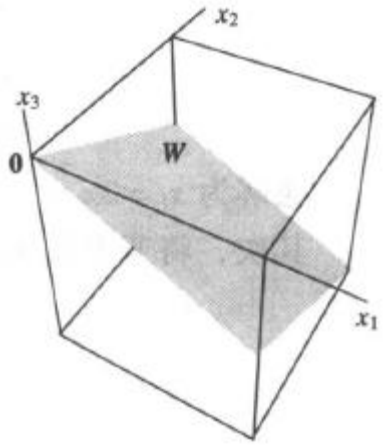
$$W = \left\{ \begin{bmatrix} 6a-b \\ a+b \\ -7a \end{bmatrix} : a,b \in \mathbb{R} \right\}$$

- Solution : First, write W as a set of linear combinations.

Let $$W = \left\{ a\begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix} + b\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} : a,b \in \mathbb{R} \right\} = \text{Span}\left\{ \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Then W = Col A.

$$A = \begin{bmatrix} 6 & -1 \\ 1 & 1 \\ -7 & 0 \end{bmatrix}$$



EXAMPLE: Find a matrix A such that $W = Col\, A$ where

$$W = \left\{ \begin{bmatrix} x-2y \\ 3y \\ x+y \end{bmatrix} : x,y \text{ in } \mathbf{R} \right\}.$$

Solution:

$$\begin{bmatrix} x-2y \\ 3y \\ x+y \end{bmatrix} = x\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + y\begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

Therefore $A = \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 1 & 1 \end{bmatrix}$.

§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

By Theorem 4 (Chapter 1),

> The column space of an $m \times n$ matrix A is all of $\mathbf{R}^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution for each $\mathbf{b}$ in $\mathbf{R}^m$.

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补充矩阵的秩(Rank)的另一个刻画条件

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一、矩阵的秩的概念

定义：在 $m\times n$ 矩阵 A 中，任取 k 行 k 列($k\le m$，$k\le n$)，位于这些行列交叉处的 k^2 个元素，不改变它们在 A 中所处的位置次序而得的 k 阶行列式，称为矩阵 A 的 k 阶子式

显然，$m\times n$ 矩阵 A 的 k 阶子式共有　　个.

概念辨析： k 阶子式、矩阵的子块、余子式、代数余子式

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix}$$

与元素a_{12}相对应的余子式

$$M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

相应的代数余子式

$$A_{12} = (-1)^{1+2} M_{12} = -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

矩阵 A 的一个 2 阶子式

$$\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

矩阵 A 的一个 2 阶子块

$$\begin{pmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{pmatrix}$$

定义： 设矩阵 A 中有一个不等于零的 r 阶子式 D，且所有

$r+1$ 阶子式（ 如果存在的话 ）全等于零，那么 D 称为矩阵

A 的最高阶非零子式，数 r 称为矩阵 A 的秩，记作

$R(A)$.

规定： 零矩阵的秩等于零.

$$A=\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix}$$

矩阵 A 的 2 阶子式

矩阵 A 的一个 3 阶子式

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

如果矩阵 A 中所有 2 阶子式都等于零，那么这个 3 阶子式也等于零 .

定义： 设矩阵 A 中有一个不等于零的 r 阶子式 D，且所有 $r+1$ 阶子式（ 如果存在的话 ）全等于零，那么 D 称为矩阵 A 的最高阶非零子式，数 r 称为矩阵 A 的秩，记作 $R(A)$.

规定： 零矩阵的秩等于零.

- 根据行列式按行（ 列 ）展开法则可知，矩阵 A 中任何一个 $r+2$ 阶子式（ 如果存在的话 ）都可以用 $r+1$ 阶子式来表示.
- 如果矩阵 A 中所有 $r+1$ 阶子式都等于零，那么所有 $r+2$ 阶子式也都等于零 .
- 事实上，所有高于 $r+1$ 阶的子式（ 如果存在的话 ）也都等于零 .

因此矩阵 A 的秩就是 A 中非零子式的最高阶数.

矩阵 A 的秩就是 A 中非零子式的最高阶数.

显然，

- 若矩阵 A 中有某个 s 阶子式不等于零，则 $R(A)\geq s$；

  若矩阵 A 中所有 t 阶子式等于零，则 $R(A)<t$.

- 若 A 为 n 阶矩阵，则 A 的 n 阶子式只有一个，即$|A|$.

  当$|A|\neq 0$ 时， $R(A)=n$;

  可逆矩阵（ 非奇异矩阵 ）又称为满秩矩阵.

  当$|A|=0$ 时， $R(A)<n$;

  不可逆矩阵（ 奇异矩阵 ）又称为降秩矩阵.

- 若 A 为 $m\times n$ 矩阵，则 $0\leq R(A)\leq \min(m, n)$.

- $R(A^{T})=R(A)$.

$$A=\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix} \qquad A^T=\begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \\ a_{14} & a_{24} & a_{34} \end{pmatrix}$$

矩阵 A 的一个 2 阶子式　　矩阵 A^{T} 的一个 2 阶子式

$$D=\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \qquad D^T=\begin{vmatrix} a_{12} & a_{22} \\ a_{13} & a_{23} \end{vmatrix}$$

A^{T} 的子式与 A 的子式对应相等，从而 $R(A^{\mathrm{T}})=R(A)$.

例： 求矩阵 A 和 B 的秩，其中

$$A=\begin{pmatrix}1 & 2 & 3\\ 2 & 3 & -5\\ 4 & 7 & 1\end{pmatrix} \qquad B=\begin{pmatrix}2 & -1 & 0 & 3 & -2\\ 0 & 3 & 1 & -2 & 5\\ 0 & 0 & 0 & 4 & -3\\ 0 & 0 & 0 & 0 & 0\end{pmatrix}$$

解： 在 A 中，2 阶子式 $\begin{vmatrix}1 & 2\\ 2 & 3\end{vmatrix}\neq 0$.

A 的 3 阶子式只有一个，即$|A|$，而且$|A|=0$，因此 $R(A)=2$.

例：求矩阵 A 和 B 的秩，其中

$$A=\begin{pmatrix}1 & 2 & 3\\ 2 & 3 & -5\\ 4 & 7 & 1\end{pmatrix}\qquad B=\begin{pmatrix}2 & -1 & 0 & 3 & -2\\ 0 & 3 & 1 & -2 & 5\\ 0 & 0 & 0 & 4 & -3\\ 0 & 0 & 0 & 0 & 0\end{pmatrix}$$

解（续）：B 是一个行阶梯形矩阵，其非零行有 3 行，因此其 4 阶子式全为零.

以非零行的第一个非零元为对角元的 3 阶子式

$$\begin{vmatrix}2 & -1 & 3\\ 0 & 3 & -2\\ 0 & 0 & 4\end{vmatrix}=24\neq 0$$，因此 $R(B)=3$.

还存在其它3 阶非零子式吗？

例：求矩阵 A 和 B 的秩，其中

$$A=\begin{pmatrix}1&2&3\\2&3&-5\\4&7&1\end{pmatrix}\qquad B=\begin{pmatrix}2&-1&0&3&-2\\0&3&1&-2&5\\0&0&0&4&-3\\0&0&0&0&0\end{pmatrix}$$

解（续）：B 还有其它 3 阶非零子式，例如

$$\begin{vmatrix}2&0&3\\0&1&-2\\0&0&4\end{vmatrix}=8\qquad \begin{vmatrix}2&-1&-2\\0&3&5\\0&0&-3\end{vmatrix}=-18\qquad \begin{vmatrix}2&0&-2\\0&1&5\\0&0&-3\end{vmatrix}=-6$$

结论：行阶梯形矩阵的秩就等于非零行的行数.



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补充线性方程组的解的判定

设有 n 个未知数 m 个方程的线性方程组

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \cdots\cdots\cdots\cdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m. \end{cases}$$



定义： 线性方程组如果有解，就称它是**相容的**；如果无解，就称它是**不相容的**.

问题1： 方程组是否有解？

问题2： 若方程组有解，则解是否唯一？

问题3： 若方程组有解且不唯一，则如何掌握解的全体？

定理： n 元线性方程组 $Ax = b$

① 无解的充分必要条件是 $R(A) < R(A, b)$；

② 有唯一解的充分必要条件是 $R(A) = R(A, b) = n$；

③ 有无限多解的充分必要条件是 $R(A) = R(A, b) < n$.

分析： 只需证明条件的充分性，即

- $R(A) < R(A, b) \Rightarrow$ 无解；
- $R(A) = R(A, b) = n \Rightarrow$ 唯一解；
- $R(A) = R(A, b) < n \Rightarrow$ 无穷多解 .

那么

✓ 无解 $\Rightarrow$ $R(A) < R(A, b)$；

✓ 唯一解 $\Rightarrow$ $R(A) = R(A, b) = n$；

✓ 无穷多解 $\Rightarrow$ $R(A) = R(A, b) < n$.

§ 4.2 Null Spaces, Column Spaces, and Linear Transformation

- The Contrast Between Nul A and Col A

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Nul A	A is a m*n matrix	Col A
1. Nul A is subspace of R^n		1. Col A is a subspace of R^m
2. Nul A is implicitly defined, that is you are given only a condition (Ax=0) that vectors in Nul A must satify		2. ColA is explicity defined; that is, you are told how to build vectors in Col A.
3. It take time to find vector in Nul A. Row operation on [A 0] are required		3. It is easy to find vectors in Col A, The columns of A are display. Others are formed from them
4. There is no obvious relation between Nul A and the entries in A		4.There is an obvious relation between Col A and the entries in A, since each columns of A is in Col A
5.A typical vector v in Nul A has the property that Av = 0		5.A typical vector v in Col A has the property that the equation Ax=v is consistent
6.Given a specific vector v, it is easy to tell if v is in Nul A, just compute Av		6. Given a specific vector v, it may take time to tell if v is in Col A. Row operation on [A v] is required.
7.Nul A={0} if and only if the equation Ax=0 has only the trivial solution		7. Col A =R^m if and only if the equation Ax=b has a solution for every b in R^m
8. Nul A={0} if and only if the linear transformation x|→Ax is one to one		8. Col A =R^m if and only if the linear transformation x|→Ax maps R^n onto R^m

Kernel（核）and Range of a Linear Transformation

A **linear transformation** T from a vector space V into a vector space W is a rule that assigns to each vector **x** in V a unique vector $T(\mathbf{x})$ in W, such that

i. $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in V;

ii. $T(c\mathbf{u}) = cT(\mathbf{u})$ for all **u** in tin V and all scalars c.

The ***kernel*** (or **null space**) of T is the set of all vectors **u** in V such that $T(\mathbf{u}) = \mathbf{0}$. The ***range*** of T is the set of all vectors in W of the form $T(\mathbf{u})$ where **u** is in V.

So if $T(\mathbf{x}) = A\mathbf{x}$, col A =range of T.

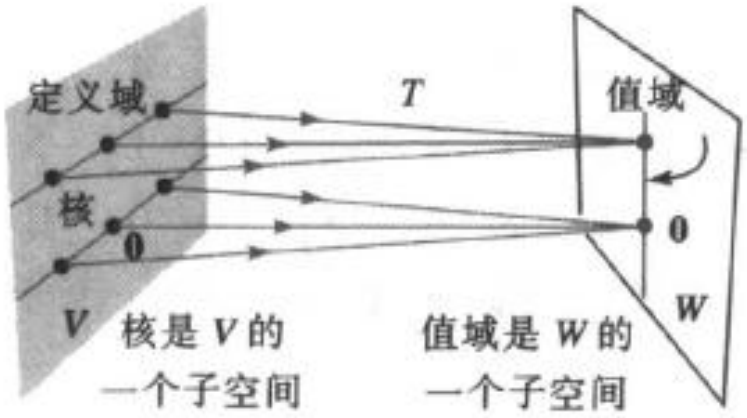


图 4-12　与线性变换相关的子空间

Example : Let V be the vector space of all real-valued functions f defined on an interval [a, b] with the property that they are differentiable(可微的) and their derivatives(导数) are continuous functions on [a, b]. Let W be the vector space of all continuous functions on [a, b] and let D: V→W be the transformation that changes f in V into its derivate f'. In calculus, two simple differentiation(微分) rules are

$$D(f+g) = D(f) + D(g) \text{ and } D(cf) = cD(f)$$

That is, D is a linear transformation. It can be show that the kernel of D is the set of **constant function** [a, b] and the range of D is the set W of all continuous functions on [a, b]

Assignment

- Ex:9,15, 30,32

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§4.3 Linearly Independent Sets; Bases

- 1. Linearly Independent Sets
- 2. The Basis set
- 3. The Spanning Set Theorem
- 4. Bases for Null A and Col A

本节定义与R^n空间中类似，但是要超越于R^n空间

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§ 4.3 Linearly Independent Sets; Bases

Linearly Independent Sets

Definition :

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ in a vector space V is said to be **linearly independent** if the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$$

has only the trivial solution $c_1 = 0, \ldots, c_p = 0$.

The set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ is said to be **linearly dependent** if there exists weights $c_1, \ldots, c_p$, not all 0, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p = \mathbf{0}.$$

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§ 4.3 Linearly Independent Sets; Bases

The following results from Section 1.7 are still true for more general vectors spaces.

A set containing the zero vector is linearly dependent.

A set of two vectors is linearly dependent if and only if one is a multiple of the other.

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§ 4.3 Linearly Independent Sets; Bases

EXAMPLE: $\left\{\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 3 & 2 \\ 3 & 0 \end{bmatrix}\right\}$ is a

linearly dependent set.

EXAMPLE: $\left\{\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 3 & 6 \\ 9 & 11 \end{bmatrix}\right\}$ is a linearly

independent set since $\begin{bmatrix} 3 & 6 \\ 9 & 11 \end{bmatrix}$ is not a multiple of $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

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Theorem 4

An indexed set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ of two or more vectors, with $\mathbf{v}_1 \neq \mathbf{0}$, is linearly dependent if and only if some vector $\mathbf{v}_j$ $(j > 1)$ is a linear combination of the preceding vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$.

Note: The main difference between linear dependence in R^n and in a general vector space is that when the vectors are not n-tuples, the homogeneous equation (1) usually cannot be written as a system of n linear equations.

EXAMPLE: Let $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ be a set of vectors in $\mathbf{P}_2$ where $\mathbf{p}_1(t) = t$, $\mathbf{p}_2(t) = t^2$, and $\mathbf{p}_3(t) = 4t + 2t^2$. Is this a linearly dependent set?

Solution: Since $\mathbf{p}_3 =$ 4 $\mathbf{p}_1 +$ 2 $\mathbf{p}_2$, $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ is a linearly dependent set.

§ 4.3 Linearly Independent Sets; Bases

- Example ： Let $p_1(t) = 1, p_2(t) = t, p_3(t) = 4 - t,$
- Example ： The set {sin t, cos t}， {sin t cos t, sin 2t}

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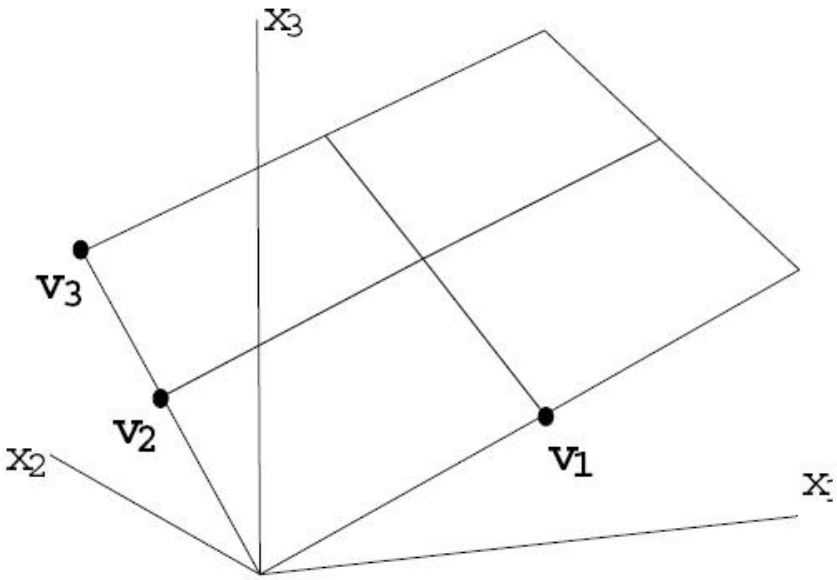
§ 4.3 Linearly Independent Sets; Bases

The Basis set

Let H be the plane illustrated below. Which of the following are valid descriptions of H?

(a) $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ (b) $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_3\}$

(c) $H = \text{Span}\{\mathbf{v}_2, \mathbf{v}_3\}$ (d) $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$



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§ 4.3 Linearly Independent Sets; Bases

- **Definition :**

Let H be a subspace of a vector space V. An indexed set of vectors $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_p\}$ in V is a basis for H if

(i) β is a linearly independent set, and

(ii) $H = \text{Span}\{\mathbf{b}_1, \ldots, \mathbf{b}_p\}$.

EXAMPLE: Let $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

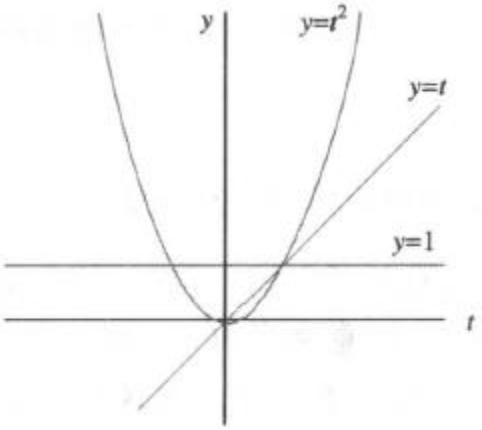
Show that $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is a basis for $\mathbf{R}^3$. The set $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is called a **standard basis** for $\mathbf{R}^3$.

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EXAMPLE: Let $S = \{1, t, t^2, \ldots, t^n\}$. Show that S is a basis for $\mathbf{P}_n$.

Solution: Any polynomial in $\mathbf{P}_n$ is in span of S. To show that S is linearly independent, assume $c_0 \cdot 1 + c_1 \cdot t + \cdots + c_n \cdot t^n = \mathbf{0}$

Then $c_0 = c_1 = \cdots = c_n = 0$. Hence S is a basis for $\mathbf{P}_n$.



The Spanning Set Theorem

A basis can be constructed from a spanning set of vectors by discarding vectors which are linear combinations of preceding vectors in the indexed set.

- Example: Let $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 6 \\ 16 \\ -5 \end{bmatrix}, H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$

Note that $\mathbf{v}_3 = 5\mathbf{v}_1 + 3\mathbf{v}_2$

Show that $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, and find a basis for the subspace H.

- Solution： Every vector in Span{v_1, v_2} belongs to *H* because

$$c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + 0\boldsymbol{v}_3$$

Now let **x** be any vector in *H*

$$\boldsymbol{x} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + c_3\boldsymbol{v}_3$$

Since $\boldsymbol{v}_3 = 5\boldsymbol{v}_1 + 3\boldsymbol{v}_2$

$$\begin{aligned}\boldsymbol{x} &= c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + c_3(5\boldsymbol{v}_1 + 3\boldsymbol{v}_2) \\ &= (c_1 + 5c_3)\boldsymbol{v}_1 + (c_2 + 3c_3)\boldsymbol{v}_2\end{aligned}$$

Thus x is in Span{v_1, v_2}, so every vector in H belongs to Span{v_1, v_2}.

{v_1, v_2} is a basis of H since {v_1, v_2} is linearly independent.

THEOREM 5 The Spanning Set Theorem

Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ be a set in V and let $H = \text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$.

a. If one of the vectors in S - say $\mathbf{v}_k$ - is a linear combination of the remaining vectors in S, then the set formed from S by removing $\mathbf{v}_k$ still spans H.

b. If $H \neq \{\mathbf{0}\}$, some subset of S is a basis for H.

Proof：

a. By rearranging the list of vector in S, if necessary, we may suppose that v_p is a linear combination of $v_1,\ldots,v_{p-1}$, say,

$$v_p = a_1v_1+\ldots+a_{p-1}v_{p-1} \qquad (3)$$

Given any x in H, we may write

$$x = c_1v_1+\ldots+c_{p-1}v_{p-1}+c_pv_p \qquad (4)$$

For suitable scalars $c_1,\ldots,c_p$. Substituting the expression for v_p from (3) into (4), it is easy to see that x is a linear combination of $v_1,...,v_{p-1}$. Thus $\{v1,\ldots,v_{p-1}\}$ spans H, because x was an arbitrary element of H.

b. If the original spanning set S is a linear independent, then it is already a basis for H. Otherwise, one of the vectors in S depends on the others and can be deleted, by part (a). So long as there two or more vectors in the spanning set, we can repeat this process until the spanning set is linear independent and hence is a basis for H. If the spanning set is eventually reduced to one vector, that vector will be nonzero (and hence linearly independent) because $H \neq \{0\}$.

Bases for Null A and Col A

(1). Bases for Null A

EXAMPLE: Find a basis for Nul A where

$$A = \begin{bmatrix} 3 & 6 & 6 & 3 & 9 \\ 6 & 12 & 13 & 0 & 3 \end{bmatrix}.$$

Solution: Row reduce $\begin{bmatrix} A & \mathbf{0} \end{bmatrix}$:

$$\begin{bmatrix} 1 & 2 & 0 & 13 & 33 & 0 \\ 0 & 0 & 1 & -6 & -15 & 0 \end{bmatrix}$$

$x_1 = -2x_2 - 13x_4 - 33x_5$

$x_3 = 6x_4 + 15x_5$

x_2, x_4 and x_5 are free

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -2x_2 - 13x_4 - 33x_5 \\ x_2 \\ 6x_4 + 15x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -13 \\ 0 \\ 6 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -33 \\ 0 \\ 15 \\ 0 \\ 1 \end{bmatrix}$$

$\uparrow$ $\uparrow$ $\uparrow$

u **v** **w**

Therefore $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a spanning set for Nul A. In the last section we observed that this set is linearly independent. Therefore $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a basis for Nul A. The technique used here always provides a linearly independent set.

(2). Bases for Col A

- Example ： Find a basis for Col B, where

$$B = [\boldsymbol{b}_1 \ \boldsymbol{b}_2 \ \cdots \ \boldsymbol{b}_5] = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- Solution： By the Spanning Set Theorem, we may discard b_2 and b_4, and $\{b_1, b_3, b_5\}$ will still span Col B. Let

$$S = \{\boldsymbol{b}_1, \boldsymbol{b}_3, \boldsymbol{b}_5\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

S is a basis for Col B.

(2). Bases for Col A

EXAMPLE: Find a basis for Col A, where

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 2 & 4 & -1 & 3 \\ 3 & 6 & 2 & 22 \\ 4 & 8 & 0 & 16 \end{bmatrix}.$$

Solution: Row reduce:

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 & \mathbf{b}_4 \end{bmatrix}$$

Note that

$\mathbf{b}_2 = \underline{2}\,\mathbf{b}_1$ and $\mathbf{a}_2 = \underline{2}\,\mathbf{a}_1$

$\mathbf{b}_4 = 4\mathbf{b}_1 + 5\mathbf{b}_3$ and $\mathbf{a}_4 = 4\mathbf{a}_1 + 5\mathbf{a}_3$

$\mathbf{b}_1$ and $\mathbf{b}_3$ are not multiples of each other

$\mathbf{a}_1$ and $\mathbf{a}_3$ are not multiples of each other

Elementary row operations on a matrix do not affect the linear dependence relations among the columns of the matrix.

Therefore Span$\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4\}$ =Span$\{\mathbf{a}_1, \mathbf{a}_3\}$ and $\{\mathbf{a}_1, \mathbf{a}_3\}$ is a basis for Col A.

§ 4.3 Linearly Independent Sets; Bases

THEOREM 6

The pivot columns of a matrix A form a basis for Col A.

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THEOREM 6

The pivot columns of a matrix A form a basis for Col A.

Proof ： The general proof uses the arguments discussed above. Let B be the reduced echelon form of A. The set of pivot columns of B is linear independent, for no vector in the set is a linear combination of vectors that precede it. Since A is row equivalent to B, the pivot columns of A are linear independent as well, because any linear dependence relation among the columns of A corresponding to a linear dependence relation among the columns of B, For this same reason, every non-pivot columns of A is a linear combination of the pivot columns of A. Thus the non-pivot columns of A may be discarded form the spanning set for Col A, by the Spanning Set Theorem. This leaves the pivot columns of A as a basis for Col A.

EXAMPLE: Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -2 \\ -4 \\ 6 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$.

Find a basis for Span$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

Solution: Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ -3 & 6 & 9 \end{bmatrix}$ and note that

Col A = Span$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

By row reduction, $A \sim \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Therefore a basis

for Span$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is $\left\{ \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} \right\}$.

$\underline{v}_1$ $\underline{v}_3$

Review:

1. To find a basis for Nul A, use elementary row operations to transform $\begin{bmatrix} A & \mathbf{0} \end{bmatrix}$ to an equivalent reduced row echelon form $\begin{bmatrix} B & \mathbf{0} \end{bmatrix}$. Use the reduced row echelon form to find parametric form of the general solution to $A\mathbf{x} = \mathbf{0}$. The vectors found in this parametric form of the general solution form a basis for Nul A.

2. A basis for Col A is formed from the pivot columns of A. **Warning: Use the pivot columns of A, not the pivot columns of B, where B is in reduced echelon form and is row equivalent to** A.

Assignment

- Ex:9,11,14,23,31

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§ 4.4 Coordinate System

- 1. Coordinate System
- 2. A Graphical Interpretation of Coordinates
- 3. Coordinates in R^n
- 4. The Coordinate Mapping

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§ 4.4 Coordinate System

Coordinate System

Theorem 7 The Unique Representation Theorem

Let $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis for a vector space V. Then for each $\mathbf{x}$ in V, there exists a unique set of scalars $c_1, \ldots, c_n$ such that

$$\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n.$$

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Proof : Since β spans V, there exist scalars such that

$x = c_1b_1+\ldots+c_nb_n$. Suppose x also has the representation

$$x = d_1b_1+\ldots+ d_nb_n$$

for scalars d1,…,dn. Then, subtracting, we have

$$0 = x-x = (c_1-d_1)b_1+\ldots+(c_n-d_n)b_n \qquad (2)$$

Since β is linearly independent, the weights in (2) must be all zero. That is , $c_j = d_j$ for $1 \leq j \leq n$.

Definition:

Suppose $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ is a basis for a vector space V and $\mathbf{x}$ is in V. The **coordinates of x relative to the basis** β (or the β – **coordinates of x**) are the weights $c_1, \ldots, c_n$ such that $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$.

In this case, the vector in $\mathbf{R}^n$

$$[\mathbf{x}]_\beta = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

is called the **coordinate vector of x** (**relative to** β), or the β – **coordinate vector of x**.

- Example: The entries in the vector $x=\begin{bmatrix}1\\6\end{bmatrix}$ are the coordinates of x relative to the standard basis $\mathcal{E}=\{e_1,e_2\}$, since

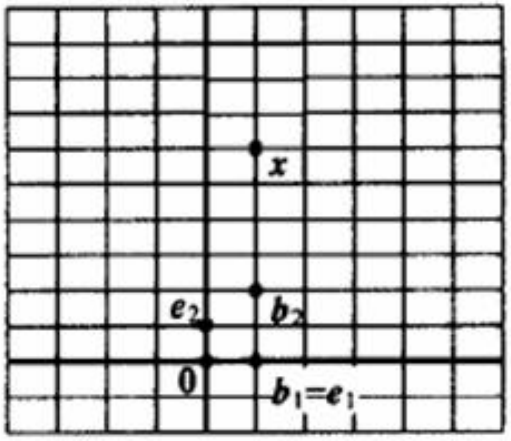
$$\begin{bmatrix}1\\6\end{bmatrix}=1\cdot\begin{bmatrix}1\\0\end{bmatrix}+6\cdot\begin{bmatrix}0\\1\end{bmatrix}=1\cdot e_1+6\cdot e_2$$

$\mathcal{E}=\{e_1,e_2\}$，则 $[x]_{\mathcal{E}}=x$

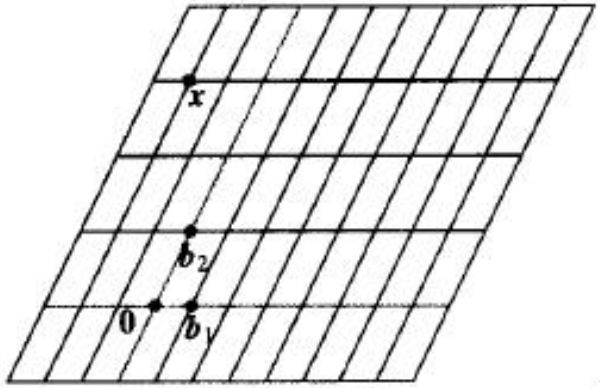
- Example: 多项式的坐标向量

A Graphical Interpretation of Coordinates

- Example： The standard basis $\mathcal{E} = \{e_1, e_2\}$, the vector $b_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, and the vector $x = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$



Standard graph paper



β – graph paper

$[x]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$

Coordinates in R^n

Example : Let $b_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, b_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, x = \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \mathcal{B} = \{b_1, b_2\}$

Find the coordinate vector $[\mathbf{x}]_\beta$ of **x** relative to **β**

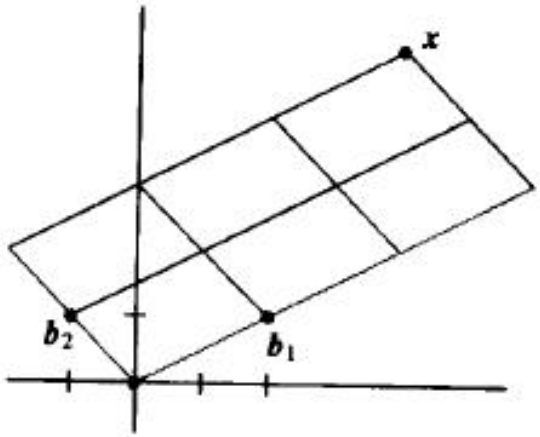
Solution : The **β**-coordinate c_1, c_2 of **x** satisfy

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$b_1 \quad b_2 \quad x$ $\qquad$ $b_1 \quad b_2 \quad x$

So $c_1 = 3, c_2 = 2$

$$x = 3b_1 + 2b_2 \qquad [x]_B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$



For a basis $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$, let

$$P_\beta = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{bmatrix} \text{ and } [\mathbf{x}]_\beta = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Then

$$\mathbf{x} = P_\beta [\mathbf{x}]_\beta.$$

We call P_β the **change-of-coordinates matrix** from β to the standard basis in $\mathbf{R}^n$. Then

$$[\mathbf{x}]_\beta = P_\beta^{-1} \mathbf{x}$$

and therefore P_β^{-1} is a **change-of-coordinates matrix** from the standard basis in $\mathbf{R}^n$ to the basis β.

The Coordinate Mapping

Coordinate mappings allow us to introduce coordinate systems for unfamiliar vector spaces.

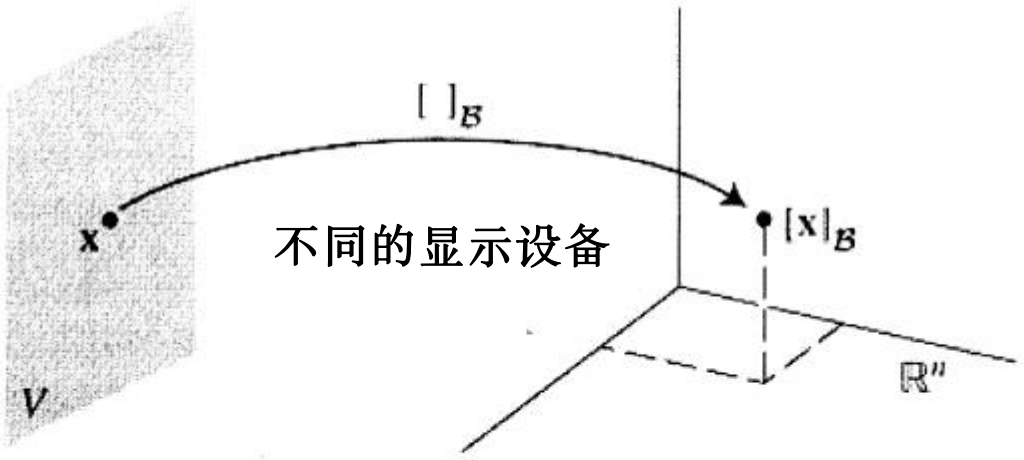


FIGURE 5 The coordinate mapping from V onto $\mathbb{R}^n$.

把不熟悉的向量空间转换为熟悉的向量空间

Theorem 8

Let $\boldsymbol{\beta} = \{b_1,\ldots,b_n\}$ be a basis for a vector space V. Then the coordinate mapping $x \mapsto [x]_B$ is a **one-to-one linear transformation** from V onto R^n.

Proof : Take two typical vectors in V

$$\boldsymbol{u} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n \qquad \boldsymbol{w} = d_1\boldsymbol{b}_1 + \cdots + d_n\boldsymbol{b}_n$$

Then $\boldsymbol{u} + \boldsymbol{w} = (c_1 + d_1)\boldsymbol{b}_1 + \cdots + (c_n + d_n)\boldsymbol{b}_n$

It follows that

$$[\boldsymbol{u} + \boldsymbol{w}]_B = \begin{bmatrix} c_1 + d_1 \\ \vdots \\ c_n + d_n \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} + \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix} = [\boldsymbol{u}]_B + [\boldsymbol{w}]_B$$

Thus the coordinate mapping preserves addition.

If **r** is any scalar, then

$$r\boldsymbol{u} = r(c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n) = (rc_1)\boldsymbol{b}_1 + \cdots + (rc_n)\boldsymbol{b}_n$$

So

$$[r\boldsymbol{u}]_{\mathcal{B}} = \begin{bmatrix} rc_1 \\ \vdots \\ rc_n \end{bmatrix} = r\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = r[\boldsymbol{u}]_{\mathcal{B}}$$

Thus the coordinate mapping also preserves scalar multiplication.

Hence it is a linear transformation.

See exercises 23 and 24 for verification that the coordinate mapping is one-to-one and maps V onto R^n.（ Assignment ）

Standard basis for $\mathbf{P}_2$: $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\} = \{1, t, t^2\}$

Polynomials in $\mathbf{P}_2$ behave like vectors in $\mathbf{R}^3$. Since
$a + bt + ct^2 =$ a $\mathbf{p}_1 +$ b $\mathbf{p}_2 +$ c $\mathbf{p}_3$,

$$[a + bt + ct^2]_\beta = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

We say that the vector space $\mathbf{R}^3$ is *isomorphic* to $\mathbf{P}_2$.

Isomorphic:同构的
Isomorphism:同构

EXAMPLE: Parallel Worlds of $\mathbf{R}^3$ and $\mathbf{P}_2$.

Vector Space $\mathbf{R}^3$	**Vector Space $\mathbf{P}_2$**
Vector Form: $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$	Vector Form: $a + bt + bt^2$
Vector Addition Example	*Vector Addition Example*
$\begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$	$(-1 + 2t - 3t^2) + (2 + 3t + 5t^2)$ $= 1 + 5t + 2t^2$

Informally, we say that vector space V is **isomorphic** to W if *every vector space calculation in V is accurately reproduced in W*, and vice versa.

EXAMPLE: Use coordinate vectors to determine if $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ is a linearly independent set, where $\mathbf{p}_1 = 1 - t$, $\mathbf{p}_2 = 2 - t + t^2$, and $\mathbf{p}_3 = 2t + 3t^2$.

把不熟悉的向量空间有关的问题转换到熟悉的向量空间上来

Solution: The standard basis set for $\mathbf{P}_2$ is $\beta = \{1, t, t^2\}$. So

$$[\mathbf{p}_1]_\beta = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, [\mathbf{p}_2]_\beta = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, [\mathbf{p}_3]_\beta = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$$

Then

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & -1 & 2 \\ 0 & 1 & 3 \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

By the IMT, $\{[\mathbf{p}_1]_\beta, [\mathbf{p}_2]_\beta, [\mathbf{p}_3]_\beta\}$ is linearly **independent** and therefore

$\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ is linearly **independent**

Assignment

- Ex4.4: 1 , 5 , 14, 23, 24

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§4.5 The Dimension of a Vector Space

What is the dimension（维度）of a vector space?

The dimension of a vector space V, denoted by *dim V,* is defined as

dim V= the number of vectors in a basis for V

is it well defined?

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§ 4.5 The Dimension of a Vector Space

- **Dimension:**

THEOREM 9

If a vector space V has a basis $\beta = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$, then any set in V containing more than n vectors must be linearly dependent.

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THEOREM 10

If a vector space V has a basis of n vectors, then every basis of V must consist of n vectors.

Proof: Suppose β_1 is a basis for V consisting of exactly n vectors. Now suppose β_2 is any other basis for V. By the definition of a basis, we know that β_1 and β_2 are both linearly independent sets.

By Theorem 9, if β_1 has more vectors than β_2, then β_1 is a linearly dependent set (which cannot be the case).

Again by Theorem 9, if β_2 has more vectors than β_1, then β_2 is a linearly dependent set (which cannot be the case).

Therefore β_2 has exactly n vectors also. ■

DEFINITION

If V is spanned by a finite set, then V is said to be **finite-dimensional**, and the **dimension** of V, written as dim V, is the number of vectors in a basis for V. The dimension of the zero vector space $\{\mathbf{0}\}$ is defined to be 0. If V is not spanned by a finite set, then V is said to be **infinite-dimensional**.

EXAMPLE: The standard basis for $\mathbf{P}_3$ is $\{\ 1, t, t^2, t^3\ \}$. So $\dim \mathbf{P}_3 =$ 4.

In general, $\dim \mathbf{P}_n = n + 1$.

EXAMPLE: The standard basis for $\mathbf{R}^n$ is $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ where $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are the columns of I_n. So, for example, $\dim \mathbf{R}^3 = 3$.

§ 4.5 The Dimension of a Vector Space

Subspaces of a Finite-Dimensional Space

THEOREM 11

Let H be a subspace of a finite-dimensional vector space V. Any linearly independent set in H can be expanded, if necessary, to a basis for H. Also, H is finite-dimensional and

$$\dim H \leq \dim V.$$

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EXAMPLE: Let $H = \text{span}\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$. Then H is a subspace of $\mathbf{R}^3$ and $\dim H < \dim \mathbf{R}^3$.

We could expand the spanning set $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}\right\}$ to $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ to form a basis for $\mathbf{R}^3$.

§ 4.5 The Dimension of a Vector Space

THEOREM 12 THE BASIS THEOREM

Let V be a p – dimensional vector space, $p \geq 1$. Any linearly independent set of exactly p vectors in V is automatically a basis for V. Any set of exactly p vectors that spans V is automatically a basis for V.

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Proof：By Theorem 11, a linearly independent set S of p elements can be extended to a basis for V. But that basis must contain exactly p elements, since dim $V = p$. So S must already be a basis for V.

Now suppose that S has p elements and spans V. Since V is nonzero, the Spanning Set Theorem implies that a subset S' of S is a basis of V. Since dim $V = p$, S' must contain p vectors. Hence $S = S'$.

EXAMPLE: Show that $\{t, 1-t, 1+t-t^2\}$ is a basis for $\mathbf{P}_2$.

Solution: Let $\mathbf{v}_1 = t, \mathbf{v}_2 = 1-t, \mathbf{v}_3 = 1+t-t^2$ and $\beta = \{1, t, t^2\}$.

Corresponding coordinate vectors

$$[\mathbf{v}_1]_\beta = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, [\mathbf{v}_2]_\beta = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, [\mathbf{v}_3]_\beta = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$\Rightarrow \{[\mathbf{v}_1]_\beta, [\mathbf{v}_2]_\beta, [\mathbf{v}_3]_\beta\}$ is linearly independent and therefore $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is also linearly independent.

Since dim $\mathbf{P}_2 = 3$, $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis for $\mathbf{P}_2$ according to The Basis Theorem.

The Dimensions of Nul A and Col A

$$\boxed{\dim \text{Col } A = \text{number of pivot columns of } A}$$

$$\boxed{\dim \text{Nul } A = \text{number of free variables of } A}.$$

EXAMPLE: Suppose $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 7 & 8 \end{bmatrix}$. Find dim Col A and dim Nul A.

Solution

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 7 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

So $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \end{bmatrix} \right\}$ is a basis for Col A and dim Col $A = 2$.

Now solve $A\mathbf{x} = \mathbf{0}$ by row-reducing the corresponding augmented matrix. Then we arrive at

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 0 \\ 2 & 4 & 7 & 8 & 0 \end{bmatrix} \sim \cdots \sim \begin{bmatrix} 1 & 2 & 0 & 4 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$x_1 = -2x_2 - 4x_4$$

$$x_3 = 0$$

and

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -4 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

So $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for Nul A and dim Nul $A = 2$.

Assigment

- Ex: 1,9

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§4.6 Rank(秩)

- 1. The Row Space
- 2. The Rank Theorem
- 3. Rank and the Invertible Matrix Theorem

- 矩阵的秩
- 向量组的秩？向量空间的秩？

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§4.6 Rank（秩）

The Row Space

The set of all linear combinations of the row vectors of a matrix A is called the **row space** of A and is denoted by Row A.

EXAMPLE: Let

$$A = \begin{bmatrix} -1 & 2 & 3 & 6 \\ 2 & -5 & -6 & -12 \\ 1 & -3 & -3 & -6 \end{bmatrix} \quad \text{and} \quad \begin{aligned} \mathbf{r}_1 &= (-1,\ 2,\ 3,\ 6) \\ \mathbf{r}_2 &= (2,-5,-6,-12) \\ \mathbf{r}_3 &= (1,-3,-3,-6) \end{aligned} .$$

$$\text{Row } A = \text{Span}\{\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3\} \text{ (a subspace of } \mathbf{R}^4)$$

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§ 4.6 Rank

THEOREM 13

If two matrices A and B are row equivalent, then their row spaces are the same. If B is in echelon form, the nonzero rows of B form a basis for the row space of A as well as B.

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- Example ： Find bases for the row space, the column space, and the null space of the matrix

$$A=\begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix}$$

Solution： Row reduce A to an echelon form

$$A \sim B=\begin{bmatrix} 1 & 3 & -5 & 1 & 5 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & -4 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Basis for Row A : $\{(1,3,-5,1,5),(0,1,-2,2,-7),(0,0,0,-4,20)\}$

Basis for Col A :

$$\left\{\begin{bmatrix} -2 \\ 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 11 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 7 \\ 5 \end{bmatrix}\right\}$$

To find Nul A, solve $A\mathbf{x} = \mathbf{0}$ first:

$$A \sim B \sim C = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 + x_3 + x_5 &= 0 \\ x_2 - 2x_3 + 3x_5 &= 0 \\ x_4 - 5x_5 &= 0 \end{aligned}$$

$x_1 = -x_3 - x_5, x_2 = 2x_3 - 3x_5, x_4 = 5x_5, x_3$ 和 x_5 为自由变量，

Basis for Nul A :

$$\left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 1 \end{bmatrix} \right\}$$

Warning：Although the first three rows of B are linearly Independent, it is wrong to conclude that the first three rows of A are linearly independent. Row operations do not preserve the linear dependence relations among the rows of a matrix.

§ 4.6 Rank

The Rank Theorem

DEFINITION

The **rank** of A is the dimension of the column space of A.

$$\boxed{\text{rank } A = \dim \text{Col } A = \#\text{ of pivot columns of } A = \dim \text{Row } A}.$$

$$\underbrace{\text{rank } A}_{\updownarrow} + \underbrace{\dim \text{Nul } A}_{\updownarrow} = \underbrace{n}_{\updownarrow}$$

$$\left\{\begin{array}{c}\#\text{ of pivot}\\ \text{columns}\\ \text{of } A\end{array}\right\} \quad \left\{\begin{array}{c}\#\text{ of nonpivot}\\ \text{columns}\\ \text{of } A\end{array}\right\} \quad \left\{\begin{array}{c}\#\text{ of}\\ \text{columns}\\ \text{of } A\end{array}\right\}$$

§ 4.6 Rank

THEOREM 14 THE RANK THEOREM

The dimensions of the column space and the row space of an $m \times n$ matrix A are equal. This common dimension, the rank of A, also equals the number of pivot positions in A and satisfies the equation

$$\text{rank } A + \dim \text{Nul } A = n.$$

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- Example : Suppose that a 5x8 matrix A has rank 5. Find dim Nul A, dim Row A and rank A^T. Is Col A = R^5?

- **Solution:**

$$\underbrace{\text{rank } A}_{\updownarrow} + \underbrace{\dim \text{Nul } A}_{\downarrow} = \underbrace{n}_{\updownarrow}$$

$$5 \qquad ? \qquad 8$$

$$5 + \dim \text{Nul } A = 8 \quad \Rightarrow \quad \dim \text{Nul } A = 3$$

$$\dim \text{Row } A = \text{rank } A = 5 \quad \Rightarrow \quad \text{rank } A^T = \text{rank } A = 5$$

Since rank *A*=# of pivots in *A*=5, there is a pivot in every row. So the columns of A span $\mathbf{R}^5$ (byTheorem 4, page 43). Hence Col *A* *Span* $\mathbf{R}^5$.

§ 4.6 Rank

THE INVERTIBLE MATRIX THEOREM (continued)

Let A be a square $n \times n$ matrix. The the following statements are equivalent:

m. The columns of A form a basis for $\mathbf{R}^n$

n. Col $A = \mathbf{R}^n$

o. dim Col $A = n$

p. rank $A = n$

q. Nul $A = \{\mathbf{0}\}$

r. dim Nul $A = 0$

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The implications (n)=>(o)=>(p) follow from the definitions Of dimension and rank. If the rank of A is n, the number of columns of A, then dim Nul A=0, by the Rank theorem, and so Nul A={0}. Thus (p)=>(r)=>(q).

Also, (q) implies that the equation Ax=0 has only the trivial solution, which is statement (d). Since statements (d) and (g) are already know to be equivalent to the statement that A is invertible.

Assignment

- Ex: 1,3

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思考题

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问题1

设有线性方程组

$$\begin{cases}(1+a)x_1+x_2+x_3=0\\ x_1+(1+a)x_2+x_3=3\\ x_1+x_2+(1+a)x_3=a\end{cases}$$

问a取何值时，此方程组

（１）有唯一解；

（２）无解；

（３）有无限多个解？并在有无限多个解时求其通解。

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- 2. 设n 元齐次线性方程组 $AX = 0$ 与$BX = 0$ 同解，证明A的秩等于B的秩。
- 3. 设A是实数矩阵。则1) 如果 $A^TA = 0$ 则$A = 0$；

  2) A^TA的秩等于A的秩。

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